Denial Appeals AI

Epic Alignment – April 6, 2026 *(stories grouped by epic — companion to the Workstream and Sprint Timeline docs)*

Project Objective

Building an AI-powered denial appeals management system that integrates with the customer's existing Azure AI Foundry instance to ingest denial documents, gather supporting evidence, generate appeal letters via the existing letter writer agent, route them through human-in-the-loop review, and deliver approved letters — accelerating appeal turnaround time and increasing overturn rates while maintaining full audit and compliance traceability.

Epic Overview

The 39 stories in the backlog are organized into 10 epics representing the major feature areas of the product. This document provides the product / feature view of the backlog — complementary to **WORKSTREAMS.pdf** (organized by capability / skill area) and **SPRINT_TIMELINE.pdf** (organized chronologically). Use this document when discussing scope with product stakeholders or mapping feature priorities.

Epic 1 Infrastructure & DevOps	**Epic 2** Authentication & Security	**Epic 3** API Proxy Layer	**Epic 4** Core App Hardening	**Epic 5** Workflow Visualization
Epic 6 HITL Queue System	**Epic 7** Denial Appeals Module	**Epic 8** Observability & Monitoring	**Epic 9** Testing & QA	**Epic 10** UX Polish & A11y

10 EPICS	**39** TOTAL STORIES	**204** TOTAL POINTS	**6** SPRINTS	**12** CALENDAR WEEKS

Epic 1 — Infrastructure & DevOps

WS1 *5 stories · 26 pts · Owner: DevOps*

Foundational Azure infrastructure, infrastructure-as-code, CI/CD automation, environment management, and networking. Everything else depends on this epic being delivered early. Sprint 1 is almost entirely Epic 1 work.

ID	Story	Description	Pts	Priority	Sprint
FV-001	**Provision Azure Resources**	Resource group, Static Web App, Key Vault, Application Insights provisioned in customer Azure tenant.	5	**Highest**	S1
FV-002	**Azure DevOps CI/CD Pipeline**	ADO Pipelines for PR validation, staging deploy on merge, manual prod approval gate via environments.	5	**Highest**	S1
FV-003	**Staging & Production Environments**	Separate staging and production resource groups with environment-specific configuration.	3	**High**	S2
FV-004	**Front Door + Custom Domain**	Azure Front Door with WAF, SSL, custom domain, rate limiting policies.	5	**High**	S5
FV-005	**Terraform / Bicep Modules**	Infrastructure-as-code modules with state management and variable-driven environments.	8	**High**	S1

Epic 2 — Authentication & Security

WS2 *4 stories · 16 pts · Owner: Backend / Security*

Identity, authentication, and authorization. Establishes Azure AD single sign-on, Microsoft Authentication Library integration, role-based access control, and credential hygiene. Customer IT approval of the app registration is the largest external dependency in this epic.

ID	Story	Description	Pts	Priority	Sprint
FV-010	**Azure AD App Registration**	App Registration with redirect URIs, API permissions for Foundry, service principal granted Azure AI Developer role.	5	**Highest**	S1
FV-011	**MSAL.js Frontend Integration**	Microsoft Authentication Library integrated for SSO, token acquisition, automatic refresh.	5	**Highest**	S2
FV-012	**Role-Based Access Control**	Admin, Reviewer, Viewer roles mapped to Azure AD groups; UI elements gated by role.	3	**High**	S4
FV-013	**Remove Hardcoded Credentials**	Security audit of source code, remove all hardcoded keys/tokens, scan history, rotate any leaked credentials.	3	**Highest**	S1

Epic 3 — API Proxy Layer

WS3 *3 stories · 16 pts · Owner: Backend*

The Azure Functions proxy that bridges the frontend to the customer's existing Foundry instance. Handles authentication translation, header forwarding, rate limiting, and request validation. This is a critical production component that solves the CORS and auth-header issues we hit during the prototype deployment.

ID	Story	Description	Pts	Priority	Sprint
FV-020	**Azure Functions API Proxy**	Functions app proxying frontend requests to Foundry; managed identity for Key Vault; CORS; logging.	8	**Highest**	S2
FV-021	**Conversations API Support**	Proxy support for conversation create + responses with bearer token forwarding for workflow identity.	5	**High**	S2
FV-022	**Rate Limiting & Validation**	Per-user rate limits, request size limits, input sanitization, auth validation on proxy.	3	**High**	S2

Epic 4 — Core Application Hardening

WS4 *5 stories · 24 pts · Owner: Frontend*

Converts the prototype React app to production quality. Replaces localStorage with proper server-side persistence, adds error boundaries, loading states, responsive design for tablets, and performance optimizations. Spread across Sprints 3-6 so hardening happens continuously as features land.

ID	Story	Description	Pts	Priority	Sprint
FV-030	**Server-Side Persistence (Cosmos DB)**	Replace localStorage with Cosmos DB for appeals, HITL items, config, execution history.	8	**High**	S3
FV-031	**Error Boundary & Global Handling**	React Error Boundaries, friendly error messages, retry options, App Insights integration, token expiry.	3	**High**	S3
FV-032	**Loading States & Skeletons**	Skeleton screens for all list/detail views; no layout shift; staggered animations.	3	Medium	S5
FV-033	**Tablet & Mobile Responsive Design**	Responsive layouts down to 768px, touch-friendly canvas, drawer navigation on small screens.	5	Medium	S6
FV-034	**Code Splitting & Performance**	Route-based lazy loading, bundle size under 300KB gzipped, Lighthouse score > 90.	5	Medium	S5

Epic 5 — Workflow Visualization

WS5 *4 stories · 21 pts · Owner: Frontend*

The signature feature of the product: live visualization of Foundry workflow execution. Includes the full YAML parser for all action types, real-time node status tracking, detailed agent configuration panels, and (as a stretch goal) side-by-side workflow comparison.

ID	Story	Description	Pts	Priority	Sprint
FV-040	**Real-Time Workflow Tracking**	Nodes animate idle → running → completed in real time as Foundry processes the workflow.	8	**High**	S3
FV-041	**Full YAML Parser Coverage**	Support for all Foundry workflow action types: Question, ConditionGroup, ForEach, GotoAction, etc.	5	**High**	S3
FV-042	**Agent Detail Configuration Panel**	Click gear icon to see full agent config: tools, instructions, model, guardrails, memory.	3	Medium	BL
FV-043	**Workflow Comparison View**	Side-by-side compare of two workflow executions with diff highlighting.	5	Low	BL

Epic 6 — HITL Queue System

WS5 *3 stories · 18 pts · Owner: Full-Stack*

The human-in-the-loop review system. Takes the workflow pause-points and turns them into a real-time, collaborative reviewer queue with conversation history, item locking, and assignment routing. This is the most operationally important feature for day-to-day use by reviewers.

ID	Story	Description	Pts	Priority	Sprint
FV-050	**Server-Side HITL Queue**	Cosmos DB or Service Bus backed queue with SignalR real-time updates and item locking.	8	**High**	S3
FV-051	**Multi-Step Conversation View**	Chat-like view of full conversation history with current question highlighted.	5	**High**	S4
FV-052	**Assignment & Routing**	Assign HITL items to specific reviewers, auto-routing, SLA tracking, escalation rules.	5	Medium	BL

Epic 7 — Denial Appeals Module

WS5 *6 stories · 36 pts · Owner: Full-Stack*

The domain-specific feature set — the reason this product exists. Connects the denial appeals management UI to the customer's existing letter writer agent via Foundry workflow integration. Includes appeal backend API, letter export, bulk processing, analytics, and evidence management. The largest epic by both story count and point total.

ID	Story	Description	Pts	Priority	Sprint
FV-060	**Appeals Backend API & Data Model**	REST API for appeal CRUD, search, pagination; full audit log; data validation.	8	**High**	S4
FV-061	**Foundry Workflow Integration**	"Process" button executes claim-dispute-workflow with real-time state tracking and HITL linking.	8	**High**	S4
FV-062	**Appeal Letter Export (PDF/DOCX)**	Formatted letter view with PDF and Word export, company letterhead, metadata.	5	**High**	S4
FV-063	**Bulk Processing**	Select multiple pending appeals, process in batch with concurrency control and progress reporting.	5	Medium	BL
FV-064	**Dashboard & Analytics**	Manager dashboard: success rate, processing time, top denial reasons, dollar value recovered.	5	Medium	BL
FV-065	**Evidence Upload & Management**	Upload supporting documents to Blob Storage; preview in app; virus scan on upload.	5	Medium	BL

Epic 8 — Observability & Monitoring

WS6 *3 stories · 13 pts · Owner: DevOps + Backend*

Telemetry, operational dashboards, and audit logging for compliance. Application Insights captures frontend and proxy behavior; Azure Monitor hosts the operations dashboard; audit logging provides the 7-year retention trail required for healthcare compliance. All three stories land in Sprint 5 just ahead of production launch.

ID	Story	Description	Pts	Priority	Sprint
FV-070	**App Insights Frontend Telemetry**	SDK initialized in React; page views, custom events, exception tracking with source maps.	3	**High**	S5
FV-071	**Operations Dashboard (Azure Monitor)**	Workbook with active users, API latency, HITL queue depth, alert rules with email/Teams delivery.	5	**High**	S5
FV-072	**Audit Logging**	Immutable audit trail of every state-changing action with 7-year retention for healthcare compliance.	5	**High**	S5

Epic 9 — Testing & QA

WS6 *3 stories · 21 pts · Owner: QA + Development*

Comprehensive test coverage including unit tests (Vitest), end-to-end tests (Playwright), and load testing (Azure Load Testing). Back-loaded into Sprint 6 so tests target stabilized features rather than moving targets. A small amount of unit test coverage accompanies each feature in earlier sprints to prevent regressions during active development.

ID	Story	Description	Pts	Priority	Sprint
FV-080	**Unit Test Suite**	Vitest + React Testing Library; 80% coverage on services/stores, 60% on components.	8	**High**	S6
FV-081	**End-to-End Test Suite**	Playwright tests for full user journey: login → process appeal → respond to HITL → export letter.	8	**High**	S6
FV-082	**Load Testing**	Azure Load Testing scenarios at 50 and 100 concurrent users; latency baseline established.	5	Medium	S6

Epic 10 — UX Polish & Accessibility

WS4 *3 stories · 13 pts · Owner: Frontend*

The finishing touches that make the product feel polished. Keyboard navigation and screen reader support for WCAG 2.1 AA compliance, in-app and browser push notifications for HITL events, and a guided onboarding experience for new users. Not critical for launch but high-impact for day-to-day usability.

ID	Story	Description	Pts	Priority	Sprint
FV-090	**Keyboard Navigation & A11y**	Full keyboard support, ARIA labels, screen reader compatibility, WCAG 2.1 AA contrast.	5	Medium	S6
FV-091	**Notification System**	In-app and browser push notifications for HITL items, completed appeals, workflow failures.	5	Medium	S6
FV-092	**Onboarding & Help System**	First-time user tour, contextual tooltips, FAQ page, "What's new" modal.	3	Low	BL

Epic Sizing & Priority Notes

- **Epic 7 (Denial Appeals Module) is the largest epic** at 36 points / 6 stories. It's also the only epic that is directly customer-facing in terms of business value — everything else is infrastructure or enabling tech. The MVP-critical stories (FV-060, FV-061, FV-062) are all in Sprint 4; the rest are deferred to backlog.
- **Epics 1 and 2 must deliver their MVP stories in Sprint 1** for the team to unblock. Everything else is blocked on infrastructure and authentication being live.
- **Epic 3 (API Proxy Layer) is small but critical.** 3 stories, 16 points, all in Sprint 2. Without this epic the frontend cannot reach Foundry in the deployed environment. We proved this during the prototype deployment when CORS and SWA auth-header stripping blocked direct calls.
- **Epic 5 (Workflow Visualization) and Epic 6 (HITL Queue) deliver the product's signature differentiators** — they are what make this a Foundry-native tool rather than a generic appeal management app. Both heavily concentrate in Sprint 3.
- **Epics 8, 9, and 10 are intentionally back-loaded** into Sprints 5 and 6. Observability, testing, and polish become more valuable as features stabilize. Front-loading them would mean testing moving targets and re-doing work.
- **7 stories across 4 epics are in the post-engagement backlog** (marked BL): FV-042, FV-043 (Epic 5), FV-052 (Epic 6), FV-063, FV-064, FV-065 (Epic 7), and FV-092 (Epic 10). Total 31 points — roughly one additional sprint of work. These are good candidates to bundle into a Phase 2 expansion after the pilot proves value.

** Sprint assignments (S1-S6, BL = Backlog) mirror the plan in SPRINT_TIMELINE.pdf. Primary workstream assignments (WS1-WS6) mirror WORKSTREAMS.pdf. Any changes should be reflected in all three documents to stay consistent.*

*** Epic owners are primary leads; in practice most stories require cross-functional collaboration. The owner is the person accountable for delivery, not the only person writing code.*

**** Story points are initial estimates and will be refined in team refinement sessions before sprint commitment.*